Finite fields and Fibonacci

Presenter notes for four 50-minute lectures

These lectures begin with the matrix method for Fibonacci periodicity, develop rings, fields, characteristic, and vector spaces, prove cyclicity of finite-field multiplication, and return to the prime-modulus period through matrix groups and a quadratic extension. The visual style follows the earlier continued-fraction lectures.

Before Lecture 1. Students should know modular arithmetic, Bezout's identity, modular inverses, 2×2 matrix multiplication and determinants, and the Fibonacci recurrence. The slides review their relevant consequences. The vector-space reminder assumes only some prior contact with linear algebra and develops span, independence, bases, coordinates, and dimension. Lecture 2 proves that dimension is well defined by the exchange argument before using dimension to count elements. Gaussian elimination is an optional alternative proof, not a prerequisite.

Before Lecture 3. Recall Lagrange's theorem for finite groups and Bezout's identity. The deck proves the polynomial root bound and the coprime-order product lemma used to construct a generator. No classification theorem for finite abelian groups or general Galois theory is used.

Before Lecture 4. Recall Fermat's little theorem and divisibility of the intermediate binomial coefficients $\binom{p}{j}$ by a prime p. The deck supplies explicit eigenvectors and proves the Frobenius argument, using Lecture 3's polynomial root bound to identify the fixed elements.

Conventions. Our rings are commutative with identity and $1 \neq 0$. A field has characteristic 0 or a prime; a finite field necessarily has prime characteristic. We use d for a vector-space dimension to avoid confusing it with an integer modulus n. We use $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and order the matrix state as $(F_{n+1}, F_n)^{\mathsf{T}}$. A restart is still described by $(F_n, F_{n+1}) = (0, 1)$. For odd p, A has determinant -1 and A^2 belongs to $\mathrm{SL}_2(\mathbb{F}_p)$.

Virtual delivery. Share the slide PDF window and keep these notes separate. Each question page precedes its answer page, so advancing the PDF reveals the answer. Give students quiet calculation time before inviting chat or spoken responses. The suggested minute allocations include that work. No animations, browser applications, or live network access are needed.

Pacing and recovery. Each main route totals 50 minutes, including its cover and exit question. Lecture 1 reaches vector spaces after 32 minutes. Lecture 2 reaches the field construction after 36 minutes. Its main slides 7–12 spend minutes 12–24 on well-definedness of dimension, including a worked exchange. Lecture 3 completes cyclicity at minute 28 and uses the next 20 minutes to count GL and SL; the exit question finishes the session. Lecture 4 reaches Frobenius at minute 29, leaving room for its proof and both period examples. These are pacing guides, not deadlines for students. If Lecture 1 runs long, shorten the recap and answer reveals, keeping student work. If Lecture 2 runs long, shorten the opening recap or the repeated pair example, keeping the exchange proof and the basis exercise. The extension-arithmetic exercise is already optional; the main route uses a worked multiplication and retains the inverse proof. In Lecture 3, the coprime-order lemma and polynomial argument are the essential new proofs; omit the general matrix count or the modulo-two exercise if students need more time. In Lecture 4, shorten the recap and the primitive-element comparison before shortening the Frobenius argument.

The optional prime-power connection. Lecture 3's extra slides compare $\mathbb{F}_{p^n}$ with $\mathbb{Z}/p^n\mathbb{Z}$. The latter is a different object: for $n > 1$ it has zero divisors, yet its unit group is cyclic when p is odd. The slides give a binomial primitive-root lifting proof and state the exact exception for powers of 2. Unit groups modulo 2 and 4 are cyclic; modulo 2^n for $n \geq 3$ they are $C_2 \times C_{2^{n-2}}$. The notes justify that classification using -1 and 5. Budget another 15–20 minutes if teaching the entire optional proof; it is outside the four 50-minute main routes.

Lecture 1: Changing the arithmetic

1. Changing the arithmetic

Minutes 0 to 1

The students already know Fibonacci periodicity via a matrix. Today names the arithmetic they have been using and begins the vector-space reminder. The goal is not diagonalization yet. Allow calculation time at slides 8 and 20. All rings in these lectures are commutative with identity, with $1 \neq 0$. A field has the additional inverse property.

2. The matrix we already know

Minutes 1 to 3

Use $F_0 = 0, F_1 = 1$. This matrix orders the state as (F_{n+1}, F_n); the book sometimes uses the reversed state and the conjugate matrix. They give the same period. Ask students to multiply A by itself. The determinant is a unit modulo every modulus, so the inverse exists. Do not confuse invertibility of this particular matrix with every nonzero residue being invertible.

3. Where are the roots?

Minutes 3 to 5

Reduce 16 modulo 11. For 7, list the squares of $0, 1, 2, 3$; the remaining inputs give the same squares. Stress that writing a square-root symbol does not manufacture a residue with that property. Division by 2 will make sense for odd primes. Prime 2 is exceptional and can wait until Lecture 3's optional material.

4. Residues are numbers with arithmetic

Minutes 5 to 7

The objects are residue classes, not just chosen representatives. Recall that replacing a by $a + kn$ changes neither the sum class nor the product class. This is the well-definedness behind modular arithmetic. Ask for the two reductions on the slide before displaying the next page.

5. The ring rules

Minutes 7 to 10

State closure under both operations. Say explicitly that $a + 0 = a$, $a + (-a) = 0$, and $a \cdot 1 = a$. Associativity allows us to omit parentheses. Integers and residue classes are the running examples; real numbers also form a ring. These are the commutative rings with identity used in the course; other conventions exist. Do not add the field inverse axiom yet. The calculations inherited from integer arithmetic justify the axioms for $\mathbb{Z}/n\mathbb{Z}$.

6. A new phenomenon: zero divisors

Minutes 10 to 12

Check both products in the last equation: 2 and 8 have the same residue modulo 6. Link cancellation failure to $2(1-4)=0$. Ask whether this could happen in the rational numbers. The point is that the arithmetic laws remain valid while familiar cancellation requires extra hypotheses.

7. Which elements can we divide by?

Minutes 12 to 14

Recall Bezout: $ax+ny=1$ supplies the inverse $[x]$. Conversely, $ax \equiv 1 \pmod{n}$ gives an integer equation $ax+ny=1$, forcing the gcd to be 1. Thus inverses are not a new computational trick: extended Euclid already computes them. A unit cannot be a zero divisor, since multiplication by its inverse would force $b=0$.

8. Your turn: modulo eight

Minutes 14 to 17

Keep this page visible while students calculate. Give a minute of quiet work, then invite answers in chat or aloud. If needed, suggest using the gcd test rather than trying every product. Ask students to reduce 3^2 and $2 \cdot 4$. The following page contains the answers.

9. Modulo eight: answers

Minutes 17 to 18

Confirm that every odd residue is coprime to 8. The nonzero even residues are zero divisors. This is also a useful reminder that an element may be its own inverse without equaling 1.

10. A field allows division

Minutes 18 to 20

Division by $a \neq 0$ means multiplication by a^{-1}. Zero has no inverse because $0b=0 \neq 1$. Integers are a ring but are not a field; for example 2 has no integer inverse. Do not claim the converse that every ring without zero divisors is a field: the integers refute it.

11. Why prime moduli give fields

Minutes 20 to 23

Give both directions. In the composite case, r cannot be invertible, since an inverse would turn $rs=0$ into $s=0$. The statement assumes $n \geq 2$. Use $p=7$ and $n=8$ as the examples students have just seen. Introduce the blackboard-bold F notation slowly: this is the same familiar residue arithmetic, now recognized as a field.

12. Characteristic: count copies of one

Minutes 23 to 25

Distinguish repeated addition from an external scalar multiplication that has not yet been defined. Characteristic zero means that no positive number of copies of 1 is zero, not that zero copies sum to zero. In residues the least positive return is exactly the modulus. Characteristic is defined for our rings as well as fields.

13. A field's positive characteristic is prime

Minutes 25 to 28

Use distributivity to explain why the product of the two repeated sums is the repeated sum with rs terms. The minimality step is essential: neither factor is zero since both indices are smaller than m. Remind students that $m = 1$ is excluded by $1 \neq 0$. The ring $\mathbb{Z}/6\mathbb{Z}$ shows why the field hypothesis is needed.

14. A finite field cannot have characteristic zero

Minutes 28 to 30

This uses finiteness and subtraction, then the preceding theorem. Do not assume the field's number of elements equals its characteristic. That tempting assumption is exactly what Lecture 2 will disprove. Here we know only that the characteristic is some positive prime.

15. Every finite field contains a prime field

Minutes 30 to 32

The map $[a] \mapsto a \cdot 1$ is well defined modulo p, respects the two operations, and is injective by minimality of the characteristic. The nonzero elements have inverses within this copy by the same Bezout argument as before. This copy is called the prime subfield. Prepare the next step: elements of this smaller field can act as scalars on K.

16. Vectors and scalars

Minutes 32 to 34

Always name both the vector space and its scalar field. Vectors need not be arrows and scalars need not be real numbers. For now use pairs to keep every operation visible. Later the vectors will be elements of a larger field. Scalar multiplication is an operation with one scalar input and one vector input, rather than two vector inputs.

17. The vector-space rules

Minutes 34 to 37

State closure under vector addition and scalar multiplication. Read each law aloud and say which symbols are scalars and which are vectors. The zero vector is distinct in type from the scalar 0, even when we use the same symbol. In coordinate spaces these axioms hold because the field laws hold in every coordinate. This is a definition, not an assertion that vectors can be multiplied together.

18. A familiar calculation over a different field

Minutes 37 to 39

Calculate each coordinate before reducing: $(3, 4)$, $(2, 4)$, and $(4, 3)$. Ask students to verify $2(u + v) = 2u + 2v = (0, 2)$. Three copies of any vector in this example sum to the zero vector. Contrast that with $\mathbb{R}^2$ without relying on a geometric picture.

19. Linear combinations and span

Minutes 39 to 42

There are three scalar choices here. The span includes zero and the original vector, and is closed under addition and scalar multiplication. For pairs $e_1 = (1, 0)$, $e_2 = (0, 1)$, every (a, b) equals $ae_1 + be_2$. Mention this as a contrast: one vector need not span the whole space. Independence and uniqueness of coordinates are Lecture 2's task.

20. Your turn: a span over $\mathbb{F}_2$

Minutes 42 to 45

Give at least a minute of quiet calculation. Encourage a four-row table for $(a, b) = (0, 0), (1, 0), (0, 1), (1, 1)$. For the third question, ask for coefficients, not merely a yes. This exercise prepares the coordinate-counting argument without introducing independence early.

21. Four coefficient choices, four vectors

Minutes 45 to 46

The table shows both spanning and uniqueness in this example. We will name the combination of those properties a basis next time. Do not generalize distinct coefficient choices giving distinct vectors to an arbitrary spanning list: independence is needed.

22. What we have built today

Minutes 46 to 48

Ask students to explain the distinction between a unit and a nonzero element. Then connect the three bullets: the copy of $\mathbb{F}_p$ supplies scalars for a vector space whose vectors are elements of the larger field. We will check its vector-space rules next time rather than asking them to accept the slogan on faith.

23. Exit question

Minutes 48 to 50

Expected answer: $2 \cdot 3 = 0$ with both factors nonzero, so 2 and 3 cannot have inverses. Our vector-space definition requires a field of scalars. Addition and scalar multiplication over a ring can still be studied; those objects are modules, an optional observation on the backup slide. End here at minute 50.

Optional slides after the 50-minute stopping point

Backup 1. Rings as scalars: a glimpse beyond vector spaces

Use only if a student asks whether the ring case is meaningless. It is meaningful, but some basis and dimension facts change. The example uses the ring itself as a module over itself. Avoid developing module theory during the main lecture.

Backup 2. One set, two different scalar fields

This example is optional because complex numbers may not be familiar. Over the reals, $1, i$ form a basis; over the complex numbers, 1 alone does. Lecture 2 supplies the precise basis definition. The example anticipates $\mathbb{F}_{49}$ over $\mathbb{F}_7$ without implying any embedding into the ordinary real or complex fields.

Lecture 2: Counting and building finite fields

1. Counting and building finite fields

Minutes 0 to 1

This lecture finishes the vector-space reminder, proves the necessary prime-power size theorem, and constructs the quadratic extension needed for Fibonacci modulo 7. The construction proves existence for this example, not for every prime power. The main route now proves that dimension is well defined by a six-slide exchange argument. Keep the basis calculation break; the extra extension-arithmetic exercise is optional material after the exit question.

2. Recall: scalars determine the combinations

Minutes 1 to 2

Retrieve rather than re-lecture: ask for $2u$, and ask how to express $(2, 1)$ in the standard coordinate vectors. A spanning list expresses every vector, but could express the same vector in different ways. The next two slides isolate that issue.

3. A redundant spanning list

Minutes 2 to 4

Point out two distinct coefficient triples for the same vector. Modulo 3, the coefficients in the relation are $1, 1, 2$, so they are not all zero. The problem is redundancy, not lack of spanning. Ask which vector can be removed without changing the span.

4. Linear independence

Minutes 4 to 7

Finish the second line verbally: independence forces every $a_i - b_i = 0$. Conversely, unique expressions of zero force independence. Give the standard pair e_1, e_2 as the basic example: $(a, b) = 0$ implies both coordinates are zero. Distinguish a nontrivial relation from one whose coefficients all vanish.

5. A basis gives existence and uniqueness

Minutes 7 to 9

Have students supply which hypothesis does which job. An ordered basis also specifies the order of coordinates. The standard basis of k^d is the list of coordinate vectors. We will count ordered bases when we count matrices in Lecture 3.

6. Build a basis one vector at a time

Minutes 9 to 12

Prove the key step. If $a_1v_1 + \cdots + a_rv_r + bw = 0$ and $b \neq 0$, then division by b expresses w in the old span, contradicting its choice. Thus $b = 0$, and old independence makes all other coefficients zero. Finiteness makes the strictly increasing sequence of spans stop. This proof specifically establishes the finite basis needed below.

7. Could two bases have different lengths?

Minutes 12 to 13

Make the logical issue explicit. Existence and uniqueness of coordinates refer to one chosen basis. They do not yet compare two different bases. The following argument works over any field, including infinite fields; it does not use the count p^d. Only finite lists are involved in the comparison theorem.

8. Exchange one vector

Minutes 13 to 15

Explain both inclusions of spans. Each vector in the new list lies in the old span. The solved equation puts the discarded w_1 in the new span, and all other old vectors remain. Hence the spans are equal. The spanning list need not be independent. Relabeling just moves a nonzero coefficient to the first position. Dividing by it is valid because the scalars form a field.

9. An exchange in $\mathbb{F}_3^2$

Minutes 15 to 17

Ask students to supply each subtraction. Over $\mathbb{F}_3$ a minus sign means adding twice that vector. Each exchange preserves the number of vectors and the span. The general proof will also need to ensure we can always remove an old w, rather than a vector already inserted from the independent list. This example anticipates the retained basis exercise later in the lecture.

10. Repeat the exchange

Minutes 17 to 20

Start with $k = 0$. Since the list spans, such an expression exists. If all coefficients of the remaining old vectors vanished, moving v_{k+1} to the other side would give a relation among the v_i with its coefficient -1, contradicting independence. Therefore solve for an old vector having nonzero coefficient and replace that vector by v_{k+1}. The earlier exchange formula preserves spanning and never removes an already inserted v_i.

11. An independent list cannot be longer

Minutes 20 to 22

The contradiction is the rigorous reason the process cannot outlast the original spanning list. If fewer than m independent vectors were given, the process stops earlier and the inequality already holds. If $m = 0$, the space is $\{0\}$ and any nonempty independent list is impossible, so the assertion remains true. The argument also shows an independent finite list can be extended to a basis when a finite spanning list is available: keep the inserted vectors and remove any remaining redundancies.

12. Dimension is well defined

Minutes 22 to 24

Only now define dimension. Such a space is called finite-dimensional. The result proves independence from the choice of basis, not independence from the choice of scalar field. The theorem compares finite bases. It also rules out an infinite independent basis when a finite spanning list exists, since an arbitrarily long finite sublist would violate the inequality. No Gaussian elimination was needed, though elimination supplies another proof of the same comparison theorem. The zero space has the empty basis and dimension zero. The next slide specializes to finite-field coordinates.

13. Count coordinate choices

Minutes 24 to 27

The coordinate map from $\mathbb{F}_p^d$ to V is onto by spanning and one-to-one by independence. Thus the product rule counts the vectors exactly, not merely bounds their number. In particular, a span of r independent vectors has p^r elements. If two bases have lengths d, e, this count gives $p^d = p^e$ and hence $d = e$.

14. Your turn: is this a basis?

Minutes 27 to 30

Give time to write $a + b = 0$ and $a + 2b = 0$. Subtract the first from the second to get $b = 0$, then $a = 0$. For $(2, 0)$, use $a + b = 2$, $a + 2b = 0$. A suggestion for students stuck in real-number habits: $-2 = 1$ in $\mathbb{F}_3$. The answer appears on the next page.

15. Basis exercise: answers

Minutes 30 to 31

There are nine elements in the ambient space, so the nine-element span is the whole space. We have checked both requirements for a basis without silently assuming that independence implies spanning in arbitrary spaces. Coordinates depend on the chosen basis: $(2, 0)$ has coordinates $(1, 1)$ in this basis.

16. A field can be a vector space

Minutes 31 to 34

Check all the vector-space axioms, not just the two displayed ones: addition in K is an abelian group operation, $(a+b)u = au + bu$, and $1u = u$. These are field laws restricted to the scalar subfield. This is not making K a vector space over the real numbers. Here a vector can also be multiplied by another vector because K has additional field structure; the vector-space definition alone does not supply that multiplication.

17. Every finite field has prime-power size

Minutes 34 to 36

Ask students to rule out 6 as a prime power. For 49, unique prime factorization gives $p = 7, d = 2$. We have proved a necessary condition; we have not proved existence or uniqueness for arbitrary p^d. The next construction supplies a particular missing example. To justify a finite basis, apply the earlier basis-building process to the finite set K. The strictly increasing spans must stop. Since $1 \neq 0$, its dimension over $\mathbb{F}_p$ is at least one. Thus finiteness first gave prime characteristic and now gives a finite basis.

18. Build the missing square root

Minutes 36 to 38

Define the elements as pairs first, so uniqueness of this representation is built in. The symbol s is not an ordinary real square root. We are defining new finite arithmetic. Write $s = (0, 1)$ and embed $a \in \mathbb{F}_7$ as $(a, 0)$. There are already 7^2 candidate elements; we still must prove this arithmetic is a field.

19. Multiply, then replace s^2 by 5

Minutes 38 to 41

Explain the four terms from distribution. This is polynomial arithmetic with the relation $s^2 = 5$; the ring laws hold. If requested, associativity can be checked from the pair formula, or justified by reducing products of polynomials modulo $X^2 - 5$. A backup slide introduces that notation. The product 2 is an element of the embedded $\mathbb{F}_7$.

20. Why every nonzero element has an inverse

Minutes 41 to 44

The second case contradicts that 5 is nonsquare in $\mathbb{F}_7$. The denominator is a nonzero element of the original field, so its inverse exists there. This proves the new ring is a field. Explain the division notation as multiplication by that inverse. If 5 were a square, the same formal construction could have nonzero zero divisors; a backup slide gives the example modulo 11.

21. A field with forty-nine elements

Minutes 44 to 46

Spanning follows from the definition as pairs, and independence follows because $a + bs = 0$ means both coordinates are zero. Seven copies of 1 sum to zero; no smaller positive number does. Thus the characteristic is 7, although the number of elements is 49. The same construction works over an odd prime with any nonsquare d replacing 5.

22. $\mathbb{F}_{49}$ and $\mathbb{Z}/49\mathbb{Z}$ are different

Minutes 46 to 48

Read the two columns slowly. The symbol 7 denotes different things in the two arithmetics: a nonzero residue in one and the sum $7 \cdot 1 = 0$ in the other. Equal cardinality is far weaker than equal arithmetic. This is the main misconception to address before moving on.

23. Exit question

Minutes 48 to 50

Expected answers: characteristic 5, dimension 2 over $\mathbb{F}_5$, and $\mathbb{Z}/25\mathbb{Z}$ has characteristic 25 and the nonzero zero divisor 5. The question assumes a field with 25 elements; it does not rely on a general finite-field existence theorem. This is the 50-minute stop.

Optional slides after the 50-minute stopping point

Backup 1. Polynomial notation for the construction

This is the polynomial analogue of integer residues. Addition and multiplication descend from polynomial arithmetic, so the ring laws follow. Polynomial division by the monic quadratic gives unique remainders of degree at most one. The main lecture proves the field property directly, so a general quotient-field theorem is unnecessary.

Backup 2. What if the chosen number is already a square?

The coefficient of s is 1 in both factors, so neither is the zero pair. This makes the nonsquare hypothesis in the inverse proof concrete. There is no need to construct an extension when the roots already lie in the original field.

Backup 3. Your turn: calculate in the new arithmetic

Allow a minute of quiet work. Ask students to show intermediate expressions so that an error in reduction can be distinguished from an error in expansion. The second calculation prepares the conjugate inverse formula and should not be described as complex conjugation.

Backup 4. Two products

Verify $6 \cdot 6 = 1 \pmod 7$. Multiplying the second identity by 6 gives the inverse of this one element. Do not yet claim that every element is invertible; the next slide proves that general statement. For a quick check, $(2 + s)(5 + s) = 15 + 7s = 1$.

Backup 5. The roots now exist

Modulo 7, $2^{-1} = 4$, so $\alpha = 4 + 4s$ and $\beta = 4 - 4s$. Their sum and product verify the factorization $(X - \alpha)(X - \beta) = X^2 - X - 1$. The roots are distinct because $s \neq 0$. This reconnects the construction to the original motivating question. Lecture 3 supplies cyclicity and matrix counts; Lecture 4 uses the roots' powers to finish the period argument.

Backup 6. A bridge to matrices

Ask students to remember how many elements are in the span of one nonzero vector over $\mathbb{F}_p$: exactly p. That is the key number for the next lecture's second column. This optional bridge previews Lecture 3; the main route already ended at the exit question.

Backup 7. Gaussian elimination gives another proof

C has m rows and r columns. Elimination produces at most m pivots, so when $r > m$ a variable is free. Set one free variable to 1, the others to 0, and solve backwards for the pivot variables. Row operations preserve the solutions, and division by a nonzero pivot is allowed over a field. The coefficients c_{ji} need not be unique: the original w_j merely span, and any chosen expressions suffice. Thus this proof does not assume the dimension theorem or rank-nullity. It is an alternative to the main exchange proof, not a prerequisite.

Lecture 3: Cyclic groups and counting matrices

1. Cyclic groups and counting matrices

Minutes 0 to 1

This lecture proves cyclicity of a finite field's multiplicative group and then counts ordered bases and determinant fibers. Recall Lagrange's theorem and the definition of element order. We prove the polynomial root bound before using it. The group-size-one case is included. The optional slides compare finite fields with units of prime-power residue rings; they are not part of the 50-minute main route.

2. The nonzero elements form a group

Minutes 1 to 3

Closure follows from absence of zero divisors: the product of two nonzero elements cannot be zero. Inverses are nonzero. Lagrange gives the displayed order divisibility. Define order as the least positive exponent returning to 1. The task is to find one element of order N, which is stronger than all orders dividing N.

3. What cyclic means

Minutes 3 to 5

Calculate the last power by reducing 8 modulo 5. A generator of a finite group of size N has order N, and conversely an element of order N supplies all N elements by its distinct powers. There may be several generators. Our aim is to prove at least one exists in every finite field, not to identify a canonical generator.

4. A polynomial cannot have too many roots

Minutes 5 to 7

Recall polynomial division by $X - a$. The constant nonzero case has no roots. Each root factors off one linear term; cancellation works since $b - a$ is a nonzero field element. This proof is the specific place field arithmetic enters the cyclicity theorem. The root bound can fail over a ring with zero divisors.

5. Build the order one prime at a time

Minutes 7 to 9

Explain the double-bar notation: ℓ^r divides N, but the next power does not. Distinct primes have relatively prime powers. The product step requires both coprime orders and commutativity, proved below. For $K = \mathbb{F}_2$, $N = 1$ and 1 is a generator of the trivial group.

6. Find an element with the required prime factor

Minutes 9 to 12

Do not assume a generator to choose a: the root bound proves that at least one of the N nonzero elements fails this equation. The chosen a can depend on ℓ, which is fine. Its order must include the full ℓ-power of N, but the next slide proves the needed statement directly through powers rather than invoking an order formula.

7. Isolate the prime-power order

Minutes 12 to 15

Read both exponent products aloud. If $x^m = 1$, its order divides m: divide m by the least positive return time and use minimality of the remainder. This fact is more precise than Lagrange and also applies here to a known exponent. The prime-power divisor observation is what forces the exact order without computing it by trial.

8. Combine two coprime orders

Minutes 15 to 19

This lemma deserves the full four minutes. If $ur + vs = 1$, then $z = z^{ur+vs} = (z^r)^u(z^s)^v = 1$; negative exponents use inverses. Hence $a^t = b^t = 1$. The order-divides-return-time fact forces both divisibilities, whose coprimality gives $rs \mid t$. We already know rs is a return time, so it is the least. Commutativity is needed for $(ab)^t = a^t b^t$. The result does not claim that arbitrary products have the lcm of their factors' orders.

9. One element of full order

Minutes 19 to 22

Check that a partial product's order remains coprime to the next prime-power order, so induction applies. An element of order N has exactly N distinct powers from exponent 0 through $N - 1$; there are only N elements in the group. The proof has used Lagrange, the polynomial root bound, Bezout, and commutativity, with no general classification theorem for finite abelian groups.

10. Primitive elements

Minutes 22 to 24

In this finite-field context primitive means generator of the multiplicative group. Such an element also generates the extension field, since a field containing it contains all its powers and zero. The converse need not hold: generating a field extension does not necessarily mean having multiplicative order $q - 1$. In particular, the Fibonacci root modulo 7 will have order 16, not 48.

11. Your turn: combine orders in $\mathbb{F}_7$

Minutes 24 to 27

Give a minute of calculation. Students should find $6^2 = 1$ with $6 \neq 1$, and $4^2 = 2$, $4^3 = 1$ with $4 \neq 1$. Coprimality lets us combine the orders. This is a concrete version of the theorem's final construction, not a search through all six possible generators.

12. A generator modulo seven

Minutes 27 to 28

The table verifies the conclusion independently. We used the lemma to know the order before listing all powers. This ends the cyclicity portion at minute 28. The rest of this lecture counts matrix groups.

13. From field elements to matrices

Minutes 28 to 30

Retrieve the coordinate count from Lecture 2. The remaining 20 minutes count GL and SL, providing the group-size bounds we will use for the Fibonacci matrix in Lecture 4. The cyclicity theorem concerns field elements under multiplication, not a claim that GL or SL is cyclic.

14. Invertible matrices and ordered bases

Minutes 30 to 32

Connect matrix invertibility to the bijectivity of its linear map. A basis supplies existence and uniqueness of solutions to $Mx = y$. Conversely, an invertible map makes the columns spanning and independent. Over a field this also agrees with nonzero determinant. Order matters: exchanging columns usually gives another matrix.

15. Choose the first column

Minutes 32 to 34

Ask students why a zero column prevents invertibility. A singleton list containing a nonzero vector is independent because a nonzero scalar has an inverse. This will matter when we count its span.

16. Choose the second column

Minutes 34 to 37

The scalar multiples are distinct because v_1 is nonzero. Being outside the span gives independence, by the basis-extension argument from Lecture 2. Two independent vectors in $\mathbb{F}_p^2$ span it: their span has p^2 elements, the full ambient size. The number of allowed second columns does not depend on which first column was chosen.

17. Your turn: count modulo two

Minutes 37 to 39

Give quiet work time, then ask for both the total count and the actual two allowed columns for this fixed choice. Students should list the four vectors and exclude 0 and v_1. This reinforces the reason for subtraction in the count.

18. Six invertible matrices

Minutes 39 to 40

Confirm that zero and the first column are exactly the forbidden vectors. Over $\mathbb{F}_2$, every nonzero determinant is 1, so GL and SL will have the same size. Mention this only as an anticipation.

19. The same count in dimension d

Minutes 40 to 42

The span of the empty list has size 1, accounting for the first factor when $r = 0$. Multiplying the independent choices counts ordered bases. Do not expand or simplify the product: the unsimplified form records the argument. Most lecture time stays with $d = 2$.

20. The determinant groups matrices by value

Minutes 42 to 44

Define $\mathbb{F}_p^\times$ as the multiplicative group of nonzero field elements, of size $p - 1$. The determinant is a surjective group homomorphism. Quotient-group notation is optional; the next slide uses a concrete bijection between determinant fibers.

21. Every determinant fiber has the same size

Minutes 44 to 46

Verify the two determinants: $\det(D_a S) = a$ and $\det(D_a^{-1} M) = a^{-1}a = 1$. Fibers are disjoint and exhaust GL. This establishes the count without assuming the first isomorphism theorem. If that theorem is familiar, the quotient is on a backup slide.

22. In dimension two

Minutes 46 to 48

Factor $p^2 - p = p(p - 1)$ before cancelling. Briefly recall the element-order definition: the least positive exponent giving the identity. Lagrange applies to the cyclic subgroup generated by an element. This assumes the course's earlier group material rather than reproving Lagrange.

23. Exit question

Minutes 48 to 50

Expected answers: $K^\times$, with multiplication, of order $q-1$. The proof finds an element of full order; many other elements can have smaller orders, including 1 when $q > 2$. Ask students also to recall the size $p(p^2-1)$ of SL(2,p). End at minute 50.

Optional slides after the 50-minute stopping point

Backup 1. The determinant quotient

This is the first isomorphism theorem. Kernel subgroups are normal, so the quotient is defined. The explicit fiber bijection in the main deck already proved the group-size formula without needing quotient terminology. This optional slide answers the literal GL/SL question.

Backup 2. A different object with a cyclic unit group

This is additional material, not an application of the finite-field theorem to a residue ring. Units are exactly the classes not divisible by p, giving $p^n - p^{n-1}$. Both factors in the displayed zero product are nonzero when $n > 1$. The next slides give a primitive-root lifting proof. The field comparison assumes a field of order p^n; we have not proved existence for every prime power in the main lectures.

Backup 3. Two arithmetics with nine elements

Calculate successive powers modulo 9. The nonzero residue 3 satisfies $3^2 = 0$. A field of nine elements can instead be constructed by adjoining a square root of 2 to $\mathbb{F}_3$, using Lecture 2's method. This contrasts a six-element unit group with an eight-element multiplicative field group, despite equal sizes of the underlying sets.

Backup 4. First lift a primitive root to modulo p^2

Choose any integer representative g prime to p. The two choices have the same primitive reduction modulo p. The linear binomial term is divisible by p but not p^2; all remaining terms are divisible by p^2. Thus both powers cannot equal 1 modulo p^2. Write $h^{p-1} = 1 + pc$ with $p \nmid c$. It would be false to say every primitive representative automatically lifts without adjustment.

Backup 5. Each further lift contributes a factor of p

Define $v_p(m)$ as the exponent of p dividing the nonzero integer m, if needed. For $2 \le j < p$, the binomial coefficient contributes p, so the jth term has valuation at least $1 + rj \ge r + 2$. The last term has valuation $rp \ge r + 2$ since $p \ge 3$. After division by p^{r+1}, the result is c modulo p, so it remains nonzero. Induction gives the displayed formula. This is a direct special-case proof, not an invocation of a general lifting-the-exponent lemma.

Backup 6. The lifted element has full order

Reduction modulo p forces $p-1$ to divide the order; Lagrange in the unit group gives the upper divisibility. The valuation formula says $h^{(p-1)p^j} \equiv 1 \pmod{p^n}$ exactly when $j+1 \geq n$. This proves the exact order, with a single adjusted integer h working for every n. The case $n=1$ is the finite-field theorem.

Backup 7. The exception starts at eight

C_m denotes a cyclic group of order m. The direct-product generators are -1 and 5. Repeated squaring gives $v_2(5^{2^k}-1) = k+2$, starting with $5-1=4$: if $u = 1+2^r c$, $r \geq 2$, c odd, then $u^2-1 = 2^{r+1}c(1+2^{r-1}c)$ and the last factor is odd. So 5 has order 2^{n-2}, and its powers are exactly the residues equal to 1 modulo 4. Every odd residue is in this subgroup or its negative, and -1 is outside it. This proves the direct product. Its largest element order is 2^{n-2}, smaller than its size 2^{n-1}, so it is not cyclic. The modulo-8 example alone suffices for a short classroom comparison.

Backup 8. Why the field proof does not apply to residue rings

The finite-field proof chose an element failing a low-degree polynomial equation. With zero divisors, cancellation fails in the factor-theorem argument. Thus the residue-ring theorem needs an additional lifting argument, and the theorem fails for powers of 2 starting at 8. This returns to Lecture 1's discussion of zero divisors.

Lecture 4: Explaining Fibonacci periods

1. Explaining Fibonacci periods

Minutes 0 to 1

The first three lectures supplied fields, coordinates, cyclic multiplicative groups, and counts of GL and SL. Today returns to Fibonacci with a full 50 minutes for the order argument and Frobenius. The prime binomial identity and Fermat's theorem are the remaining prerequisites. Keep the calculation break before the modulo-11 example.

2. The structures we now understand

Minutes 1 to 3

Recall that a primitive element has order $q - 1$; not every element is primitive. Matrix groups need not be cyclic. The determinant of A is -1, so its square rather than A itself lies in SL. Our field-group theorem is about multiplication of nonzero scalars.

3. The Fibonacci period is the matrix order

Minutes 3 to 7

The displayed power formula follows by induction from the recurrence, and is the students' existing matrix approach. The equivalence needs both state coordinates, not merely $F_t = 0$. If a student asks why an orbit period should equal the whole matrix order, this identity is the specific Fibonacci justification; that equality need not hold for an arbitrary matrix and arbitrary starting vector.

4. A first bound from SL_2

Minutes 7 to 9

For an element of even order t, the least positive r with $A^{2r} = I$ is $t/2$, because t must divide $2r$. This proves the order identity. The broad group-size divisibility uses no splitting information. Its purpose is to motivate exploiting the special matrix, rather than presenting the group size as an exact Fibonacci period.

5. Choose a field containing the roots

Minutes 9 to 12

In the split case, s is an existing square root, not a new symbol. In the nonsplit case, the proof from Lecture 2 generalizes directly from 7 to any odd p: $a^2 - 5b^2$ cannot vanish for a nonzero pair. The prime 5 is excluded because the roots coincide, and 2 because division by 2 is unavailable. Both roots are nonzero since their product is -1.

6. Diagonalize in the field where it makes sense

Minutes 12 to 16

Compute the product $(r+1, r)^{\mathsf{T}}$ and use the root equation. This explicitly proves the eigenvectors and invertibility of the eigenvector matrix, so no general diagonalization theorem is needed. Changing the field lets us find P; it does not change whether a power of the original matrix equals I. Its entries still lie in the embedded prime field. This also gives the lcm of the two root orders.

7. A period is a simultaneous order

Minutes 16 to 18

An element's return times are exactly the multiples of its order, by division with remainder. We need both eigenvalues to return to 1. This lcm statement is about simultaneous return, not the order of the product of arbitrary elements. The coprime-product lemma from Lecture 3 had an additional coprimality hypothesis.

8. Case one: the roots are in $\mathbb{F}_p$

Minutes 18 to 21

Remind students that the roots are nonzero, so Fermat applies. The conclusion is divisibility, not equality. The new bound is much sharper than the broad SL group-size bound because we have used the eigenvalues of this particular matrix.

9. Your turn: roots modulo eleven

Minutes 21 to 24

Give students a minute to calculate before discussing. Expected roots are 8 and 4; $8^2 = 9$, $8^4 = 4$, $8^5 = -1$. These rule out both 2 and 5 as return times, and therefore all proper divisors of 10. The next page reveals the answers and exact period.

10. Modulo eleven: an exact period

Minutes 24 to 26

Compute $2^{-1} = 6$ modulo 11, giving roots $5 \cdot 6 = 8$ and $(-3) \cdot 6 = 4$. To check the powers, $8^2 = 9$, $8^4 = 4$, and $8^5 = 32 = -1$. If A^5 or A^2 were I, the corresponding power of 8 would be 1. We have ruled out every proper divisor of the known multiple 10. The order of the other root need not be computed.

11. A Fibonacci root need not be primitive

Minutes 26 to 29

This comparison connects cyclicity directly to the order problem. In the field from Lecture 2, $\alpha^2 = 5 + 4s$, $\alpha^4 = 5s$, and $\alpha^8 = (5s)^2 = 125 = -1$. Thus its exact order is 16. Nevertheless $s = 2\alpha - 1$, so α does generate this field over $\mathbb{F}_7$. Generating the extension field and generating its multiplicative group are different requirements. Frobenius will explain the order bound without needing these special calculations.

12. A new symmetry: Frobenius

Minutes 29 to 33

Recall why p divides $\binom{p}{j}$ for $0 < j < p$: the numerator contains a factor p, while $j!$ is prime to p. The power map preserves the field operations and is a bijection, so it is a field automorphism, called Frobenius. A nonzero field element cannot have a zero power. Finiteness is needed for the injective-to-surjective step; no infinite-field claim is required.

13. Which elements does Frobenius fix?

Minutes 33 to 36

For nonzero prime-field elements use Fermat; zero is immediate. Recall the polynomial root bound proved in Lecture 3. The polynomial is nonzero since its leading coefficient is 1, even in characteristic p. This argument identifies the fixed elements in our extension without assuming the desired conjugation rule.

14. Case two: Frobenius exchanges the roots

Minutes 36 to 39

The factorization $X^2 - 5 = (X - s)(X + s)$ in K and absence of zero divisors give exactly those two square roots. They are distinct since p is odd and $s \neq 0$. If $s^p = s$, the preceding fixed-element result would put s in the prime field, contradicting nonsquareness. Frobenius fixes 1 and 2^{-1}, so applying it to the formulas for α, β interchanges them.

15. The minus sign tells us the return time

Minutes 39 to 42

Conjugating the scalar matrix $-I$ leaves it unchanged. Since the characteristic is odd, $-I \neq I$. Applying the matrix to $(1, 0)$ gives $(F_{p+2}, F_{p+1}) = (-1, 0)$, hence the displayed state in the usual order. This connects the minus sign to the earlier warning that a Fibonacci zero alone is not a restart.

16. Modulo seven: the other exact period

Minutes 42 to 45

This exact-order argument avoids a long power calculation in the extension. Ask students why $A^8 \neq I$ rules out every proper divisor of 16. For a direct Frobenius check in the field constructed last time, $s^7 = s(s^2)^3 = 125s = -s$. The general proof explains why this calculation is not a coincidence.

17. What the algebra explained

Minutes 45 to 47

Recap the distinction between existence of a period, a known multiple, and the least period. The matrix construction proves existence for all moduli; group structure and the root field give increasingly specific information for prime moduli. Prime 29 is an optional example where the bound is not attained.

18. Exit question

Minutes 47 to 50

Expected answer: a restart requires $(F_{p+1}, F_{p+2}) = (0, 1)$, but the state is $(0, -1)$ because $A^{p+1} = -I$. The period divides $2(p + 1)$ and does not divide $p + 1$; it need not equal $2(p + 1)$. End the main lecture here at minute 50. Use optional slides only if time remains or as material for a subsequent meeting.

Optional slides after the 50-minute stopping point

Backup 1. Why the prime five is different

Compute $A - 3I$ modulo 5 to get the displayed nonzero nilpotent matrix. The off-diagonal entry in A^t forces $t = 0$ in the field, hence $5 \mid t$, because 3 is nonzero. Then $A^t = 3^t I$ and 3 has order 4 modulo 5: $3^2 = -1$ and $3^4 = 1$. These two divisibilities are also sufficient, giving least time 20. The repeated root produces a nilpotent term instead of two independent eigenvectors.

Backup 2. The prime two

The short state cycle gives the exact answer immediately. This also explains why the odd-characteristic parity argument and the root formula with denominator 2 were explicitly restricted.

Backup 3. A bound need not be the least period

Check $6^7 = -1$ by repeated squaring: $6^2 = 7$, $6^4 = 20$, $6^7 = 20 \cdot 7 \cdot 6 = 28$. Thus 6 has order 14, since every proper divisor of 14 divides 7 or 2. The other root is -6^{-1}, so $24^7 = (-1)^7 (6^7)^{-1} = 1$; since 7 is prime and $24 \neq 1$, its order is 7. This is a concrete proper-divisor example, not just a caution about the word divides.

Backup 4. An exact-period procedure

Matrix repeated squaring performs the tests efficiently. At every stage the true order divides t. If t were still larger than the order, some prime factor of $t/\pi(p)$ could be removed while leaving a multiple of the order. Thus the stopping rule is exact. For 29, start at 28 and remove one factor 2 to reach 14.

Course connections and scope

The primary materials are the local number-theory course book: `chapters/modular_fibonacci.tex` for reversible states and period definitions, `chapters/finite_fields.tex` for quadratic extension arithmetic, Frobenius, and Fibonacci root orders, and `chapters/matrix_algebra.tex` for the matrix toolkit. The group and binomial chapters supply the prerequisites for Lectures 3 and 4. The ring and vector-space development and the ordered-basis count expand that route for these lectures.

The lectures prove that every finite field has prime-power order. They construct the quadratic fields required when 5 is nonsquare over an odd prime field. They prove cyclicity of finite-field multiplicative groups, using Lagrange, the polynomial root bound, and a coprime-order product lemma. They do not prove existence and uniqueness of finite fields of every prime-power size or quadratic reciprocity. Those classification results are not needed for the displayed Fibonacci divisibilities. The exact-period examples use only powers and known multiples of the order.